

KOSUMPHISAI, Thailand

WMO: 483820

Lat: 16.245N

Lon: 103.069E

Elev: 506

StdP: 14.43

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 58.5	(c) 61.1	(d) 50.0	(e) 54.4	(f) 65.8	(g) 52.7	(h) 60.2	(i) 66.3	(j) 14.5	(k) 77.0	(l) 13.7	(m) 77.1	(n) 2.5	(o) 40	(p) 0.333

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 18.4	(c) 101.4	(d) 79.3	(e) 99.3	(f) 79.3	(g) 97.3	(h) 79.2	(i) 82.8	(j) 94.4	(k) 82.1	(l) 93.0	(m) 81.5	(n) 92.0	(o) 4.9	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.2	159.7	87.6	79.6	156.7	87.0	79.0	153.5	86.4	47.4	94.6	46.5	93.3	45.8	92.0	92.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.2	12.1	9.6		DB	54.1	103.8	2.7	2.1	52.2	105.3	50.6	106.5	49.1	107.7	47.1	109.3
				WB	51.3	85.9	2.9	2.5	49.2	87.7	47.5	89.2	45.9	90.6	43.8	92.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.5	75.6	80.2	84.3	87.4	86.5	86.0	84.6	83.7	83.2	82.4	80.0	76.2	(6)
	DBStd	5.19	4.58	5.19	5.21	4.08	3.01	2.45	2.66	2.16	2.06	2.36	3.74	4.70	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	1	1	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	11866	795	845	1064	1121	1130	1080	1072	1045	997	1003	901	813	(10)
	CDD65	6393	331	425	600	671	665	630	607	580	547	538	451	348	(11)
	CDH74	73593	2947	4606	7473	8979	8437	8028	7198	6536	5997	5751	4462	3179	(12)
	CDH80	34058	1152	2245	4023	5080	4286	3893	3143	2553	2254	2293	1890	1246	(13)
(14) Wind	WSAvg	2.9	3.2	2.8	2.9	2.8	2.8	3.4	3.6	3.3	2.2	2.3	3.0	3.1	(14)
(15) Precipitation	PrecAvg	51.2	0.4	0.6	1.8	3.4	6.7	6.6	7.6	9.3	10.3	3.7	0.7	0.1	(15)
	PrecMax	66.8	2.9	2.6	5.2	9.6	13.8	11.8	13.3	13.7	16.6	8.7	4.3	0.8	(16)
	PrecMin	39.2	0.0	0.0	0.0	0.8	1.8	2.6	0.9	2.6	2.8	0.4	0.0	0.0	(17)
	PrecStd	6.7	0.6	0.7	1.5	2.3	2.9	2.5	3.1	2.6	3.5	2.0	1.0	0.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.3	99.3	103.1	104.8	102.8	99.0	98.3	95.0	94.1	93.9	95.4	94.3	(19)
		MCWB	74.6	76.3	77.0	80.5	82.1	80.8	80.1	80.9	81.2	80.0	78.5	77.1	(20)
	2%	DB	91.5	96.6	100.5	102.2	99.2	96.3	95.2	92.9	92.3	92.1	93.2	91.6	(21)
		MCWB	74.9	76.4	77.8	79.5	80.9	80.5	80.1	80.5	80.7	79.3	78.0	76.0	(22)
	5%	DB	89.1	94.3	98.1	99.7	96.7	94.6	93.0	91.4	90.8	90.7	91.2	89.2	(23)
		MCWB	74.4	76.5	78.1	79.8	80.6	80.3	79.8	80.1	80.3	78.5	77.0	74.6	(24)
	10%	DB	86.5	91.7	95.6	97.2	94.3	92.8	91.1	89.7	89.3	89.5	89.3	86.9	(25)
		MCWB	72.9	75.8	78.0	79.5	80.3	80.1	79.6	79.8	79.8	77.9	76.3	73.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.9	80.8	82.5	83.6	84.2	83.6	82.9	82.5	82.7	81.8	81.4	79.7	(27)
		MCDB	89.8	92.8	96.3	98.8	98.1	93.8	92.8	91.6	90.9	90.6	92.5	91.0	(28)
	2%	WB	77.3	79.3	80.7	82.3	82.7	82.3	81.4	81.5	81.8	80.8	79.8	77.6	(29)
		MCDB	87.4	91.4	94.4	96.9	95.0	92.2	91.1	90.4	89.7	89.4	90.0	87.9	(30)
	5%	WB	75.8	78.1	79.6	81.3	81.8	81.4	80.5	80.8	81.0	80.0	78.7	76.2	(31)
		MCDB	85.4	90.1	93.5	95.5	93.1	91.0	90.0	89.3	88.6	88.1	88.0	86.4	(32)
	10%	WB	74.2	76.9	78.7	80.3	81.0	80.7	79.9	80.1	80.4	79.2	77.4	74.7	(33)
		MCDB	83.9	88.6	92.5	94.1	91.5	90.1	88.7	87.9	87.5	86.8	85.8	84.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.7	21.7	19.8	18.4	15.6	13.8	12.8	12.1	12.1	13.9	17.9	20.6	(35)
		MCDBR	23.8	23.8	22.9	21.0	18.6	16.2	15.7	14.4	14.2	15.3	19.4	21.7	(36)
	5% WB	MCWBR	9.9	8.3	6.9	6.0	5.7	4.7	4.6	5.0	5.2	5.3	6.5	8.3	(37)
		MCWBR	20.9	20.9	20.4	19.6	16.8	14.8	14.1	13.6	13.3	14.2	17.5	20.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.479	0.592	0.658	0.618	0.513	0.465	0.453	0.462	0.505	0.535	0.477	0.471	(40)
	taud		2.074	1.796	1.688	1.821	2.139	2.294	2.326	2.292	2.157	2.036	2.170	2.121	(41)
	Ebn at Noon		255	232	221	231	253	263	266	267	256	244	255	254	(42)
	Edh at Noon		50	69	79	69	50	42	41	43	49	54	45	47	(43)
(44) All-Sky Solar Radiation	RadAvg		1551	1646	1681	1816	1757	1647	1540	1502	1469	1576	1560	1505	(44)
	RadStd		91	80	81	76	94	82	108	94	100	110	57	38	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (22 neighbors)	+0.52	N/A	N/A	N/A	+0.29	+0.23	N/A	N/A	N/A	+191

Nomenclature: See separate page